

NIKHIL ASHOK MULGIR

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SUMMARY

Machine Learning Engineer and Data Professional experienced in building scalable Python data pipelines and production-grade ETL systems, evidenced by architecting an end-to-end AWS document extraction platform and automating enterprise SQL workflows in regulated environments. Delivered ML-integrated platforms using RAG-based vector search, fine-tuned transformers, and cross-environment collaboration to ensure reliable, high-availability data solutions.

TECHNICAL SKILLS

- **Cloud & DevOps:** DevOps, AWS, Google Cloud Platform
- **Programming:** Python, SQL, R, Bash, Spark, MATLAB, PowerShell
- **Data Engineering & Analytics:** NumPy, Pandas, Spark, PySpark, Hadoop, Hive, Statsmodels, Databricks, Socket.io
- **Databases:** MySQL, PostgreSQL, BigQuery, MongoDB, DynamoDB, Redis
- **Machine Learning:** Scikit-learn, PyTorch, TensorFlow, Keras, XGBoost
- **Visualization & Tools:** TIBCO DV Studio, Matplotlib, Seaborn, Plotly, Tableau, Power BI, Git, GitHub, Linux, BitBucket, Jira, JFrog, Microsoft Office

PROFESSIONAL EXPERIENCE

Programmer Analyst – Cognizant Technology Solutions, Pune

Sep 2022 - Jul 2024

- Designed and Automated optimized SQL procedures and ETL workflows for 50+ Oracle tables using TIBCO Data Virtualization and SQL scripts, reducing query time by 30% and identifying bottlenecks to improve system uptime by 15%.
- Directed the secure handling and ETL of patient data across Dev, QA, and Prod environments, ensuring 100% compliance while efficiently routing data to application frontends via RESTful APIs and OData endpoints.
- Orchestrated the deployment of SQL procedures using Jenkins Pipeline, JFrog, and BitBucket, and Led 20+ code reviews and unit testing sessions to achieve a 100% deployment success rate.
- Collaborated with cross-functional teams to streamline release cycles for Novartis Pharma's healthcare operations, cutting deployment time by 20% while ensuring zero downtime in production systems.

Intern – Cognizant Technology Solutions, Pune

Aug 2021 - Sep 2022

- Engineered and optimized 20+ complex SQL queries and foundational ETL scripts during technical simulations, improving data extraction efficiency by 20% for mock enterprise datasets.
- Developed prototype data pipelines utilizing Python and RESTful APIs, successfully integrating 3 disparate data sources and reducing manual processing time by 25% within agile training sprints.

PROJECTS

VERDICT – AI-Powered Deposition Coaching & Trial Preparation Platform

- Spearheaded a 4-agent AI orchestration system by fine-tuning NVIDIA Nemotron-8B for contradiction confidence scoring, achieving $\leq 1.5s$ objection alert latency across 20+ concurrent witness sessions.
- Unified a RAG pipeline over a vector-indexed legal corpus with Databricks Delta Live Tables, AWS S3, and Socket.io WebSockets into a Next.js/Fastify, enabling real-time semantic inconsistency detection and Witness Composure Timelines.
- Assessed ElevenLabs STT/TTS voice pipeline quality using Galileo evals, benchmarking transcription accuracy and response coherence across Claude SDK and Conversational AI integrations, achieving 87% transcription accuracy.

Data Extraction & ML Pipeline for BASCULE (Capstone Project), UMD College Park

- Architected an end-to-end document extraction pipeline using Python and PyTorch, fine-tuning a Donut model to achieve a 91% F1-score, and benchmarked against XGBoost baselines to optimize entity-level accuracy and reduce false positives.
- Operationalized a fault-tolerant AWS architecture (S3, Lambda, DynamoDB) using PySpark to reduce latency by 90%, integrating Power BI for model monitoring and Git-based CI/CD workflows to ensure production reliability.

HACKATHON AWARDS

- [Startup Week Buildathon, New York](#) Feb 2026
Tech: Multi-Agent Systems, LLM Fine-Tuning, RAG, Vector Search, ElevenLabs, Databricks, Model Evals, Galileo
- [Technica - T. Rowe Price Investor Challenge, College Park](#) Dec 2025
Tech: Vector Embeddings, NLP Sentiment Modeling, LLM Fine-Tuning, Semantic Search, RAG
- [ServiceNow AI Challenge, Orlando](#) Nov 2025
Tech: Sentence Transformers, Clustering, Multi-Agent Systems, Agent Orchestration, Model Context Protocol

EDUCATION

Robert H. Smith School of Business, University of Maryland, College Park

Expected May 2026

Master of Science in Information Systems

GPA 3.93/4.0

- Terrapin Scholarship Recipient
- Coursework: Data Mining, Designing AI Systems, Data Analysis using Python, Data Models and Statistics using R, DBMS

Vishwakarma Institute of Technology, Pune

Aug 2018 - Jun 2022

Bachelor of Technology

GPA 3.3/4.0

- Patent Granted : 'System for Automatic Detection of Quality of Filter of Air Purifier' (4 claims)